# Topic 5: Discovering Requirements (part 2)

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
Universiti Teknologi Malaysia



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01 WHAT, HOW AND WHY?

02 WHAT ARE REQUIREMENTS?

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# WHAT, HOW, & WHY?

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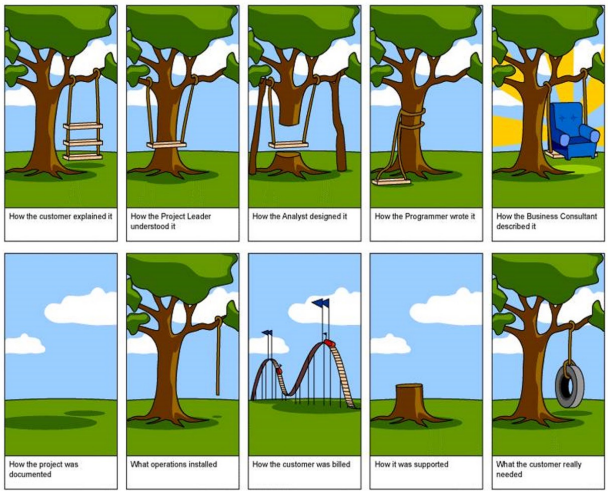
## What, How?

- What is the purpose of the requirements activity?
  - Explore the problem space
  - Establish the design challenge to be addressed
- How can requirements be captured once discovered
  - In prototypes or operational product
  - Through structured or rigorous notations
  - Criteria to show when requirements are fulfilled
  - Different physical and digital representations emphasise and de-emphasise different aspects

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# Why are requirements important?




Requirements activity is the stage where miscommunication occurs most commonly

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# WHAT ARE REQUIREMENTS?

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# What are Requirements?

- A statement about an intended product that specifies what it is expected to do or how it will perform
- Different forms and different levels of abstraction
- User stories (prevalent in agile development contexts)
- Format:

  As a <role>, I want <behavior> so that <benefit>
- Example user stories for a travel organiser might be:

  As a <traveler>, I want <to save my favorite airline for all my flights> so that <I will be able to collect air miles>

  As a <travel agent>, I want <my special discount rates to be displayed to me> so that <I can offer my clients competitive rates>

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# Volere Shell

Requirement #: **75** Requirement Type: **9** Event/use case #: **6**

Description: **The product shall issue an alert if a weather station fails to transmit readings.**

Rationale: **Failure to transmit readings might indicate that the weather station is faulty and needs maintenance, and that the data used to predict freezing roads may be incomplete.**

Source: **Road Engineers**
Fit Criterion: **For each weather station the product shall communicate to the user when the recorded number of each type of reading per hour is not within the manufacturer's specified range of the expected number of readings per hour.**

Customer Satisfaction: **3** Customer Dissatisfaction: **5**
Dependencies: **None** Conflicts: **None**
Supporting Materials: **Specification of Rosa Weather Station**
History: **Raised by GBS, 28 July**

Volere
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# Different kinds of Requirements

- Functional:
  - What the system should do
- Data:
  - What kinds of data need to be stored?
  - What are the characteristics of the data?



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# Different kinds of Requirements (cont.)

- Environment or context of use:
  - **Physical**: dusty? noisy? vibration? light? heat? humidity? …. (for example, in a hospital)
  - **Social:** collaboration and co-ordination, data sharing, distributed, synchronous or asynchronous, privacy
  - **Support:** user support, communications structure and infrastructure, availability of training
  - **Technical:** on which technologies will it run or with which it needs to be compatible?



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# Different kinds of Requirements (cont.)

- User characteristics — Who are they?
  - **Characteristics:** educational background, personal circumstances, abilities or skills
  - **System use**: novice, expert, casual, frequent
    - **Novice**: prompted, constrained, clear
    - **Expert**: flexibility, access/power
    - **Frequent:** shortcuts
    - **Casual/infrequent**: clear menu paths
  - User profile



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# Different kinds of Requirements (cont.)

- Usability goals
- User experience goals
- Different products have different requirements and may be implemented in different ways, for example, security



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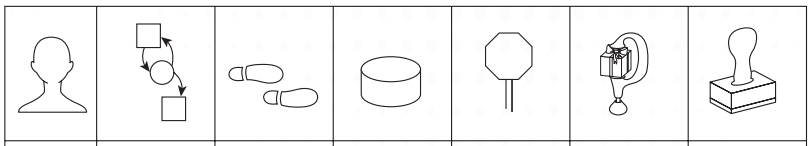
# Usable Security

- How to make security robust without detracting from user experience
  - If the usability of security is ignored, then security mechanisms will be circumvented
  - What might affect usable security
    - Making security mechanisms visible or not
    - Sonification of password strength
    - Good design may have undesirable effect on security



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# The Seven Product Dimensions



| User | Interface | Action | Data | Control | Environment | Quality Attribute |
|---|---|---|---|---|---|---|
| Users interact with the product | The product connects to users, systems, and devices | The product provides capabilities for users | The product includes a repository of data and useful information | The product enforces constraints | The product conforms to physical properties and technology platforms | The product has certain properties that qualify its operation and development |

Source: Gottesdiener and Gorman (2012), p.58. Used courtesy of Ellen Gottesdiener



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# DATA GATHERING FOR REQUIREMENTS



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## Data Gathering for Requirements

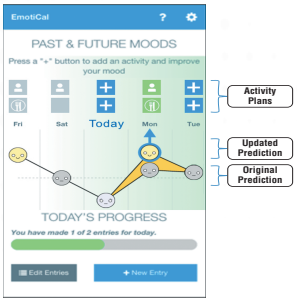
- Interviews, observation, and questionnaires
- Studying documentation:
  - Procedures and rules are often written down in manuals
  - Good source of data about the steps involved in an activity and any regulations governing a task
  - Not to be used in isolation
  - Good for understanding legislation and getting background information
  - No stakeholder time, which is a limiting factor for other techniques
- Researching similar products:
  - Good for prompting requirements



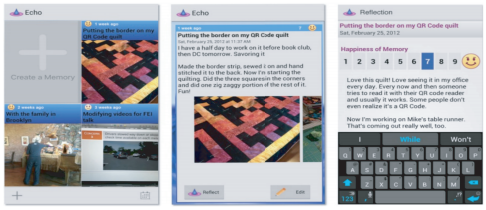
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# Combining Data Gathering

Direct observation, indirect observation, interviews, diaries, and surveys




(a)




(b)

Source: Hollis et al (2017), Figure 1.
Used courtesy of Taylor and Francis





# Combining Data Gathering

- *Diaries and interviews*: multiple information devices
- *Interviews, think aloud evaluation, questionnaire, evaluation of working prototype*: memory aid for traumatic brain injury
- *Studying documentation, evaluating other systems, user observation, and focus groups*: ship's maneuvering system
- *Questionnaire, focus group, design probe, and user study*: smart meters for power consumption and privacy

# Using Probes with Stakeholders

- Many types of probe:
  - Designed to prompt stakeholders into action
  - For researchers to learn about participants
  - All rely on data logging: automatic or manual
- Design probe:
  - Form relates specifically to particular question and context
- Cultural probe:
  - Participants asked to answer questions using postcards, maps, camera, photo album, and diary
- Technology probe:
  - A working prototype placed in an in-the-wild context
- Provocative probe:
  - Designed to challenge existing norms and attitudes





# Contextual Inquiry

- Part of Contextual Design, and used on its own to discover requirements
- One-on-one field interviews (contextual interviews)
  - 1.5 to 2 hours long
  - Focus on daily life at home or work relevant to the project
  - Uses a model of master (participant) and apprentice (researcher)
- Four main principles
  - **Context:** Going to the participant, wherever they are, and seeing what they do as they do it
  - **Partnership:** Participant & interviewer explore participant's life together
  - **Interpretation:** Observations interpreted by participant & interviewer together
  - **Focus:** Project focus guides what should be paid attention to

# Contextual Inquiry (cont.)

- Interview guided by "cool concepts" divided into two groups
- Joy of life concepts:
  - How products make our lives richer and more fulfilling
  - Accomplish, connection, identity, and sensation
- Joy of use concepts:
  - Describe impact of using the product
  - Direct in action, the hassle factor, and the learning delta
- Interview in four parts
  - Overview, transition, main interview, and wrap-up
- Following interview, interpretation session
  - Contextual design models are created or consolidated
  - Most relevant models are chosen by team, out of 10 suggested





# Brainstorming for Innovation

- Alex Osborn's (1930s) four rules:
  - Quantity over quality
  - Criticisms should be withheld
  - Encourage out-of-the-box thinking
  - Combine, refine and improve ideas

- Brainstorming requires careful facilitation

# BRINGING REQUIREMENTS TO LIFE



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## Bringing Requirements to Life

- Augmenting the basic requirements expressed as stories, in Volere template, or in other form
- Personas
  - Characterises someone who might use the product, not specific people
- Scenarios
  - An informal narrative story, simple, 'natural', personal, and not generalisable



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# Scenarios and Personas

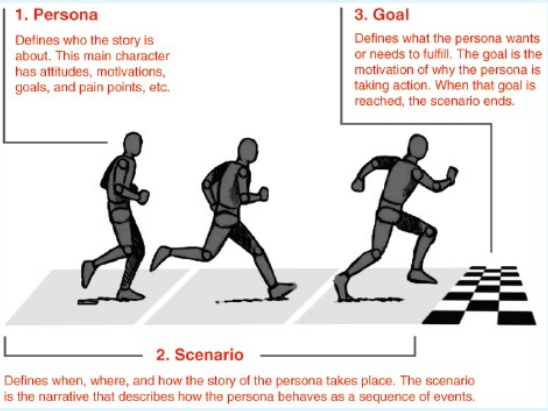



**Figure 10.10** The relationship between a scenario and its associated persona
*Source:* http://www.smashingmagazine.com/2014/08/06/a-closer-look-at-personas-part-1/



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# Personas

- Capture a set of user characteristics (user profile)
- Synthesised from real people based on research
- Typical, not idealised
- Bring to life with name, characteristics, goals, and personal background
  - Relevant to product under development
- Good persona helps designer with design decisions and reminds team about who will use the product
- Develop a small set of personas with one primary



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# Example Persona #1

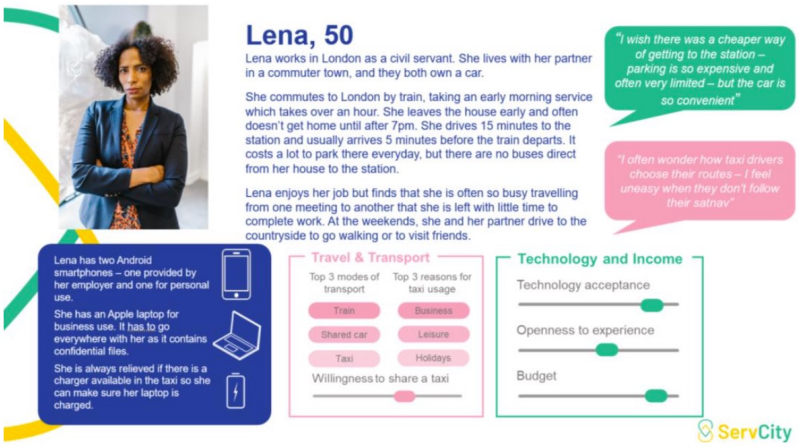
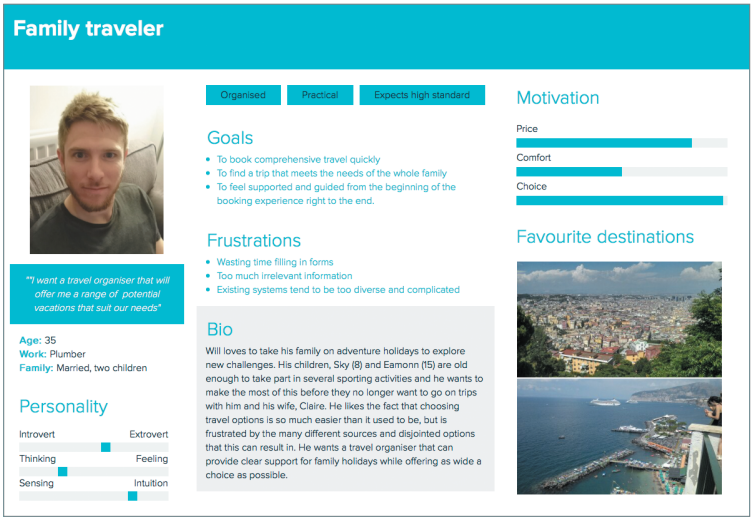



**Figure 5.** Example persona "Lena." Photo by RODNAE Productions from Pexels.



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# Example Persona #2




Developed using Xtensio Templates



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# Scenario for Group Travel Organiser

"The Thomson family enjoy outdoor activities and want to try their hand at sailing this year. There are four family members: Sky (8 years old), Eamonn (12 years old), Claire (32), and Will (35). One evening after dinner they decide to start exploring the possibilities. They want to discuss the options together but Claire has to visit her elderly mother so will be joining the conversation from her mother's house down the road. As a starting point, Will enters an idea they had been discussing over dinner – a sailing trip for four novices in the Mediterranean. The system supports users to log on from different locations and use different devices so that all members of the family can interact easily and comfortably with it wherever they are. The system's initial suggestion is a flotilla, where several crews (with various levels of experience) sail together on separate boats. Sky and Eamonn aren't very happy at the idea of going on vacation with a group of other people, even though the Thomson's would have their own boat. The travel organizer shows them descriptions of flotillas from other children their ages and they are all very positive, so eventually, everyone agrees to explore flotilla opportunities. Will confirms this recommendation and asks for detailed options. As it's getting late, he asks for the details to be saved so everyone can consider them tomorrow. The travel organizer messages them a summary of the different options available."

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# Scenarios

- May be textual descriptions, animations, audio or video
- Example animation scenarios




Source: Tommy Nilsson, et al (2020) Visions, Values, and Videos: Revisiting Envisionings in Service of UbiComp Design for the Home, DIS '20: Pages 827–839
ACM Publications.

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# Design Fiction

- Communicate a vision with future technologies
- Fictional world in which ethics, emotions, and context can be explored without concrete constraints
- Examples:
  - Privacy and surveillance
  - Exploring ethics
- Scenarios are about "overcoming the monster," while design fiction is about "quest"



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# CAPTURING INTERACTION USING USE CASES

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# Use Cases

- Focus on functional requirements and capture interaction
- Can be used in design or to capture requirements
- Use cases are step-by-step descriptions of interactions
- Two styles:
  - Essential use cases: division of tasks, no implementation detail
  - Use case with normal and alternative courses: more detail







# Example Essential Use Case for Travel Organiser

retrieveVisa

| USER INTENTION | SYSTEM RESPONSIBILITY |
|---|---|
| Find visa requirements | |
| | Request destination and nationality |
| Supply required information | |
| | Obtain appropriate visa info |
| Obtain copy of visa info | |
| | Offer info in different formats |
| Choose suitable format | |
| | Provide info in chosen format |

Note: The user intention and system responsibility are offset vertically, showing a sequence of interactions





# Use Case for Travel Organiser

1. The product asks for the name of the destination country
2. The user provides the country's name
3. The product checks that the country is valid
4. The product asks the user for their nationality
5. The user provides their nationality
6. The product checks the visa requirements of that country for a passport holder of the user's nationality
7. The product provides the visa requirements
8. The product asks whether the user wants to share the visa requirements on social media
9. The user provides appropriate social media information





# Alternative Courses for Travel Organiser

Some alternative courses:

4. If the country name is invalid:
    - 4.1: The product provides an error message
    - 4.2: The product returns to step 1
6. If the nationality is invalid:
    - 6.1: The product provides an error message
    - 6.2: The product returns to step 4
7. If no information about visa requirements is found:
    - 7.1: The product provides a suitable message
    - 7.2: The product returns to step 1

# Summary

- A requirement is a statement about an intended product that specifies what it is expected to do or how it will perform.
- Articulating requirements avoids miscommunication, supports technical developers and allows stakeholders to contribute.
- Different kinds of requirements: functional, data, environmental (context of use), user characteristics, usability goals, and user experience goals.
- Scenarios are a story-based narrative to explore existing behavior, potential of new products, and futuristic visions of use.
- Personas capture realistic profiles of people who might use the product and are synthesized from data gathering.
- Scenarios and personas can be used together throughout the product lifecycle.
- Use cases capture details about an existing or imagined interaction between users and the product.